

Project Scope Document

Project: Clean & Protect Co. – Mobile application development scope (PHASE – 1)



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1. Introduction & Context

Clean & Protect Co. aims to digitally transform its janitorial operations by introducing a centralized mobile-first platform to replace paper-based tracking, manual inspections, and reactive issue management.

Phase 1 focuses exclusively on **mobile application development**, forming the operational backbone for subsequent web and analytical modules. The application will empower **Technicians, Team Leads**, and **Clients** with real-time visibility, accountability, and proactive communication

2. Project Objective

To deliver a **mobile application (Android & iOS)** that:

- Enables field technicians to record daily cleaning activities and upload proof.
- Allows team leads to assign, inspect, and verify service quality.
- Provides clients with visibility into service progress, feedback submission, and SLA-driven ticket closure.
- Creates a unified digital ecosystem for real-time accountability and transparent service reporting.

3. Scope Overview

3.1 In-Scope (Phase 1)

The Phase-1 delivery focuses on a mobile-first operational platform for field execution, inspections, SLA workflows, and client feedback, supported by a lightweight Admin Console for core configuration.

Mobile Application (Technician, Team Lead, Client roles)

- Role-based dashboards for task execution, inspections, and service visibility.

- Technician site check-in via **QR code, site access code, or GPS geo-verification**.
- Optional **geo-targeting** on specific tasks: tasks can only be viewed/started/completed when the technician is physically at the location.
- Team Lead capability for **bulk task assignment and reassignment**.
- Client ability to submit structured feedback, including numeric + descriptive ratings and media attachments.
- **Sub-ticket workflow** for repeat issues instead of reopening closed tickets.

Contracts / Work Orders (Phase-1 Foundation)

- Operations Admin creates **Contracts/Work Orders** defining:
 - Client & site(s)
 - Service category & frequency
 - Assigned inspection template
 - Geo-targeting requirement
 - SLA rule (Standard 5-day / Critical 24-hour)
- All tasks, inspections, and SLA workflows are tied to these contracts to ensure data traceability.

Admin Console (v1) — Controlled Access (Operations Admin / Senior Admin)

- Configure **geo-targeting, SLA thresholds, and site check-in methods** (QR / Code / GPS).
- Create and manage **Contracts/Work Orders**.
- Manage **site geofences**, coordinates, and site-level settings.
- **Form Builder & Template Publisher** for digital inspection and feedback templates (derived from client's Site Visit, Monthly Check-In, and Branch forms).
- Publish templates instantly to the mobile app without requiring app updates.
- Access audit logs for configuration changes.

Core Mobile Modules

- **Task & Schedule Management:** Contract-based task generation, daily/weekly scheduling, bulk assignment, self-assignment after check-in.
- **Inspection & Quality Scoring:** Template-driven inspections, photo proof, 1–5 scoring scale.
 - **SLA automation:**
 - Score ≤ 2 → **24-hour SLA**
 - Score = 3 → **5-day SLA**
- **SLA Ticketing:** Auto-created tickets, countdown timers, alerts at 50% and 100%.
- **Client Feedback:** Hybrid rating scale (numeric + descriptive), sub-ticketing for repeat issues.
- **Notifications:** Real-time updates for assignments, reassignments, SLA events.
- **Audit Logs:** Comprehensive action logging for compliance

3.2 Out of Scope (Deferred to Future Phases)

Item / Feature	Description	Planned Phase
Operations Web Portal	Full supervisor/admin dashboards, SLA configuration, KPI overview.	Phase 2
Client Relations Dashboard	Multi-site visibility, communications tracking, and follow-ups.	Phase 3
KPI & Analytics Layer	Reports, dashboards, and audit exports for management.	Phase 4
SaaS Enablement	Multi-tenant model for white labelling to other janitorial firms.	Phase 4
Payment Gateway & Subscription Management	For future SaaS revenue model.	Phase 4

4. Functional Overview

4.1. User Authentication & Role Access

- Secure login using Firebase Authentication.
- Role-based access ensuring that each user only sees the functions relevant to their role (Technician, Team Lead, Client).
- Admin-level configuration is accessible only through the Admin Console by Operations Admin / Senior Admin.

4.2. Task & Schedule Management

- Tasks are generated under an **active Contract / Work Order**, created by Operations Admin.
- Team Leads can assign or **bulk-assign** tasks using a Quick Assign interface.
- Tasks display daily/weekly schedules with filters for “To-Do,” “In Progress,” and “Completed.”
- Technicians must perform **mandatory site check-in** before viewing or completing tasks:
 - QR Code scan
 - Site Access Code
 - GPS geo-verification
- **Optional geo-targeting** can be enabled per task.
 - When enabled, technicians cannot *view*, *start*, or *complete* the task until they are physically within the configured geofence.
- Technicians may **self-assign** tasks that are unassigned at a location once they check in.

4.3. Inspection & Quality Scoring

- Team Leads conduct inspections using templates published by the Admin Console via the **Form Builder & Template Publisher**.
- Templates support multi-field inputs:
 - Text, dropdowns, checkboxes, photo upload, rating scales.
- All inspections use a **1–5 rating scale** with mandatory comments/photos when required.

Automated SLA Logic:

- Score < 4 → corrective ticket auto generated.
- Score **3** → 5-day SLA applies.
- Score ≤ 2 → **critical SLA** with **24-hour response requirement**.

4.4. Corrective Action & SLA Workflow

- Tickets are auto created based on inspection scores.
- SLA countdown is tracked in real time with 50% and 100% alerts.
- Team Leads and Operations Admins can add corrective proof (photos, comments).
- Repeat issues automatically generate **sub-tickets**, allowing issue recurrence tracking per site.

4.5. Client Feedback & Rating

- Clients can submit feedback through a structured feedback form.
- The feedback system uses a **hybrid scale**:
 - Numeric rating (1–5)
 - **Descriptive labels**: Satisfactory, Somewhat Satisfactory, Unsatisfactory
- Optional photo/video attachments allowed.
- If the client reports a repeat issue, instead of reopening the original ticket, the system creates a **sub-ticket** linked to the main ticket for recurrence tracking.

4.6. Notifications & Audit Logs

- Real-time push notifications for:
 - New tasks
 - Reassignments
 - SLA warnings (50%, 100%)
 - Ticket updates
- Complete offline-first capability:
 - Tasks, inspections, and photos queued when offline

- Auto-sync when connection returns
- Every user action (check-in, completion, assignment, SLA update) is captured in the audit log for compliance and traceability.

4.7. Contracts / Work Orders

Before tasks can be created or assigned, the **Operations Admin / Senior Admin** must define a **Contract or Work Order**, containing:

- Client name and associated sites
- Service category (Daily Clean, Deep Clean, Compliance etc.)
- Service frequency (Daily/Weekly/Monthly/Ad-hoc)
- Assigned inspection template
- SLA rule (Standard 5 days / Critical 24 hrs)
- Geo-targeting toggle
- Effective start/end date

All Phase-1 tasks originate from Contracts/Work Orders to maintain traceability and ensure service alignment

5. User Roles and Responsibilities

Role	Key Responsibilities	Access Level
Technician	Execute cleaning tasks, check in via QR, access code, or GPS , upload photo proof, mark completion, and view tasks available after check-in	Mobile only
Team Lead	Assign/reassign tasks, bulk assign, inspect work, record quality scores, and initiate SLA tickets.	Mobile + Web-lite
Client / Customer	View service status, submit feedback, rate service with descriptive labels, and raise repeat-issue sub-tickets.	Mobile / Web-lite
Operations Admin	Configure Contracts/Work Orders, set geo-targeting rules, manage check-in logic, SLA thresholds, and publish templates	Admin Console
Senior Admin	Review and approve configurations, manage	Admin Console

template publishing, monitor compliance.

6. User Stories & Acceptance Criteria

6.1 Technician

ID	User Story	Acceptance Criteria
T1	As a Technician, I can log into the mobile app using my credentials and see only the sites assigned to me.	Technician can access only sites where they have checked in using QR, site code, or GPS . Technician can self-assign unassigned tasks after check-in
T2	As a Technician, I can view daily and weekly task lists with status, due time, and checklist.	Tasks grouped by day; “To-Do”, “In Progress”, “Completed” filters active.
T3	As a Technician, I can mark a task complete and upload photo evidence of the work.	Upload under 5 MB; success confirmation visible; sync queue enabled offline.
T4	As a Technician, I can receive push notifications when a new task is assigned or reopened.	Notification arrives ≤ 30s of event; opens correct task record.
T5	As a Technician, I can view historical tasks (30 days) for reference.	Scrollable log sorted by date.

6.2 Team Lead

ID	User Story	Acceptance Criteria
L1	As a Team Lead, I can assign and reassign daily tasks to technicians.	Task appears instantly on technician dashboard; updated on backend.
L2	As a Lead, I can conduct inspections using a 1–5 rating scale and add remarks/photos.	System prevents submission without rating and photo if required.
L3	As a Lead, when a score < 4 is submitted, the system automatically generates a corrective ticket with SLA countdown (5 days).	Ticket appears in SLA dashboard; due date calculated automatically.
L4	As a Lead, I can view ticket status and receive alerts before SLA expiry.	50% and 100% SLA notifications received.
L5	As a Lead, I can view summary reports of team performance.	Report views by technician/site with scores, tasks, and SLA metrics.

6.3 Client

ID	User Story	Acceptance Criteria
C1	As a Client, I can log in and view my site's ongoing and completed cleaning activities.	Dashboard lists current tasks, completion % and timestamps.
C2	As a Client, I can submit a feedback rating (1–5) with optional text and photo/video attachments.	Form validates rating; upload success prompt shown.
C3	As a Client, if dissatisfied, I can reopen a closed ticket.	Ticket status changes to “Reopened” and linked to same case ID.
C4	As a Client, I can view SLA progress for open issues.	SLA bar and remaining days shown dynamically.

7. UI/UX Design Expectations

- **Technician View:** Simple cards for each task, large “Complete” button, offline indicator, upload proof icon.
- **Lead View:** Task overview grid, “Inspect” action opens checklist modal, SLA tab for pending tickets.
- **Client View:** Service dashboard with “Rate Service” button, visual indicator for SLA performance.
- **Color Coding:** Green (Completed), Yellow (Pending), Red (Overdue).
- **Accessibility:** Font scaling & simple icons for non-technical field staff.

8. Data Flow Overview

1. **Task Flow:**
Supervisor (future phase) → Team Lead → Technician → Completion & photo → Sync to Cloud.
2. **Inspection Flow:**
Team Lead inspects task → Enters score → If <4 → Auto SLA Ticket → Timer begins → Corrective proof → Closure confirmation.
3. **Feedback Flow:**
Client → Submits rating/photo → Auto alert to Operations Admin → If <4 → Corrective flow triggered

9. Dependencies

Client-Side

- Provision of inspection templates and task forms by category (Retail, Healthcare, Industrial).
- Brand guidelines, logo, and colour palette.
- Site and user master data (Technicians, Leads, Clients).

- Access to current partner app for UX benchmarking.

Technical

- Firebase setup (Auth + FCM).
- Storage bucket access (AWS/Azure).
- API gateway configuration.

10. Deliverables

Stage	Deliverable	Description
Design	UI/UX Mockups	High-fidelity screens for all roles.
Development	Mobile App Build (v1.0)	Technician, Lead, and Client modules implemented.
Integration	API + Storage	REST APIs integrated for task, feedback, and SLA.
QA/UAT	Test Cases + Report	Functional, regression, and sync testing completed.
Deployment	Release Build	Production-ready Android/iOS builds deployed.
Documentation	Handover Kit	User guide, admin manual, API schema, training deck.

11. Risks & Mitigation

Risk	Impact	Mitigation
Delay in client template delivery	Medium	Use generic checklists, update dynamically post-UAT.
Technician tech literacy	High	Simplified UI, icon-based workflows, tutorial screens.
Connectivity drops	Medium	Local queue + background sync.
Security compliance with photos	Medium	Encryption at rest, signed URLs for retrieval.

12. KPIs

Metric	Target Value	Frequency
Task completion within SLA	≥ 90%	Weekly
Avg. inspection score	≥ 4.2	Monthly
Client satisfaction index	≥ 4.0	Monthly
Proof upload success rate	≥ 95%	Daily
App uptime	≥ 99%	Continuous

13. Phase Roadmap Summary

Phase	Objective	Primary Modules
Phase 1 (Current)	Mobile App Foundation	Technician, Lead, Client apps, task, feedback, SLA, audit
Phase 2	Operations Web Portal	Supervisor & Admin dashboards, SLA setup, reporting
Phase 3	Client Transparency & Relations	Client dashboard, ticket workflows, feedback analytics
Phase 4	Analytics & SaaS Expansion	KPI dashboards, subscription management, payments

14. Approval

Name	Designation	Signature	Date
Gerald/Kenna	CEO/Executive - Clean & Protect Co		
Lenin Pappachen	Vice president, PPTS Solutions		